Order Flow under a One-Position Constraint

Signal mechanics and conditional economics of a selected NQ portfolio

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October 2026

Abstract

How do heterogeneous order-flow signals interact when only one futures position may be held? We examine an existing fifteen-cell Nasdaq-100 futures portfolio, separating structural continuation, rejection, absorption, and range-reclaim mechanisms from the allocation rule that determines which opportunities are owned. Over 279 admitted sessions ending September 11, 2026, the included historical path records 1,098 accepted trades, USD115,849 net PnL, a 1.54 profit factor, and USD5,658 maximum daily-close drawdown. Attribution is conditional on ownership, not standalone performance or a sleeve-removal effect. The last five and ten admitted sessions are negative despite positive full-history and thirty-session totals. Documented timing and selection limitations prevent interpreting these results as a proven causal online strategy or untouched out-of-sample evidence.

Keywords: order flow; futures; volume profiles; portfolio capacity; historical simulation.
JEL: G11, G14, C58. **Data cutoff:** September 11, 2026.

1 The economic question

A price level is a hypothesis about where trading behavior may change, not a trading signal by itself. A burst of aggressive buying at an upper boundary may indicate acceptance of higher prices; the same buying, if it makes little progress and is followed by an inward move, may indicate rejection. This study examines an existing portfolio that distinguishes these responses across volume profiles, time-at-price structures, rolling extremes, and a volume-weighted price envelope.

Allocation is equally important. The fifteen cells share one NQ position; an accepted opportunity can block another. A cell's contribution depends on its timing, earlier owners' holding periods, and the surviving candidate population. Adding independently traded component PnLs does not reconstruct this portfolio.

We explain the signal primitives, information clocks, payoff geometry, and conditional attribution of this existing selection. August 7, 2025–September 11, 2026 contains 1,264 included candidate rows, 1,098 accepted trades, and 166 overlap rejections. No strategy, threshold, roster, or allocation rule is changed.

Evidence scope. This is descriptive analysis of an inspected, selected historical simulation, not independent raw-data replication. Known issues concern Completed-21 source availability, retrospective overnight close marks, and VWAP bar-eligibility, validation, and close-handling differences described below. Their lifetime incidence and economic effects remain unreconciled.

They qualify causal interpretation; they do not establish a measured amount or direction of PnL bias. We preserve the recorded path rather than silently substituting a cleaner execution model.

2 Structural levels and the profile-response engine

2.1 What is being measured?

NQ is the E-mini Nasdaq-100 futures contract: one index point is USD20 and the minimum 0.25-point tick is USD5. Historical identities are exact contract-symbol/instrument pairs, not a current contract hardcoded across the sample. Regular trading hours (RTH) are expressed in `America/Chicago`, with an 08:30 start and a 15:00 close boundary.

The input is trade-event data with displayed best bid and offer sampled at trades (TBBO), not a full queue or continuous depth history. For trade j, let price be p_j, size v_j, receive time t_j, and signed aggressive volume a_j (positive for buying, negative for selling). A completed 500-contract window W_k has

$$V_k = \sum_{j \in W_k} v_j = 500, \qquad \Delta_k = \sum_{j \in W_k} a_j. \tag{1}$$

Boundary trades may be split deterministically for volume-window construction. The availability of a completed window is distinct from its internal price observations. Two confirmations must have distinct physical completion events; splitting a large trade must not manufacture two independent observations.

An anchored volume profile (AVP) describes traded volume by price relative to its source anchor. A time-price-opportunity (TPO) profile describes price occupancy over time brackets. VAL and VAH are lower and upper value-area boundaries; VPOC is the volume point of control. These are different structural references even when their numerical prices coincide. An outward direction $d \in \{-1, +1\}$ points through the approached boundary; the rejection direction is $-d$. Table 1 specifies the principal constructions and their units from the source materials [7].

2.2 Acceptance and rejection are different state transitions

The shared accepted-break primitive requires two adjacent fresh 500-contract windows. The first overlaps the level and closes beyond its full zone; both signed deltas and closes point outward, without an intervening return to the broken zone boundary. First-interaction pullback continuation additionally waits at least five seconds, observes a pullback no more than four ticks outside the broken edge without reentering the zone, then requires a fresh outward window closing at least four ticks beyond it. Second-interaction cells require completed rearm rather than counting another touch in the same interaction.

For flow-flip reversals, let D_1 denote signed flow in the current one-second interval and D_{-1} the preceding one-to-two-second interval. The geometric flow score is

$$F = \sqrt{|D_{-1} D_1|}, \qquad \operatorname{sgn}(D_{-1}) = d, \quad \operatorname{sgn}(D_1) = -d, \quad F \geq 25. \tag{2}$$

The geometric mean penalizes a large burst paired with negligible opposing flow. First-interaction flow-flip reentry also requires a preceding accepted break and later full-zone reentry. Thus an opposite trade alone does not constitute rejection.

Table 1: Structural construction and interaction conventions. All clocks are Chicago time; one tick is 0.25 NQ index point.

Structure	Source definition used in the study
Overnight D2	Prior calendar-day 17:00 to session-day 08:30, end exclusive; qualifying trades must also be received before 08:30. Sum deduplicated trade volume in 0.25-point bins. VPOC is the maximum-volume bin, lower price on a tie. Expand contiguously from it toward the larger adjacent volume until at least 70% is covered, choosing the lower adjacent bin on ties; endpoints are VAL/VAH. Require all 31 half-hour event and trade buckets and one exact contract pair.
Rolling-30	Prior 30 supplied completed one-minute RTH bars, with opens in [08:30,15:00); shift before rolling so the signal bar is excluded. Excursion [4,16) and reclaim [1,4) are *index points*, not ticks.
AVP source anchors	ETH: prior 17:00, frozen at 08:30. RTH: 08:30, refreshed hourly at 09:30–14:30. New balance: first two-close traversal of a frozen 08:30 21-session edge; anchor at second-close availability, first snapshot at least 30 minutes later. Only completed, receive-available one-minute bars enter. The primary profile allocates each bar's volume uniformly over its one-tick price range; overlapping AVP-S/M zones use the narrowest representative. This is a bar-volume proxy, unlike the exact overnight profile.
Full zone	Frozen interval $[\ell, u]$ from the source's lower/upper tick bounds converted to points. An upward accepted break closes strictly above u; a downward break below ℓ. Subsequent *full-zone* reversal reentry means a trade at or below ℓ after an upward break, or at or above u after a downward break.
Profile rearm	After a touch, five consecutive supplied completed one-minute bars must close at least eight ticks from the zone, with combined deduplicated volume at least 500. A nearer close resets both counters. Rearm is recorded at bar availability; interaction two requires that record by its own touch availability. Two consecutive closes traversing the opposite boundary retire the interaction sequence instead.

Table 2: The ten profile/flow cells. Gates are seconds; stop and target are NQ points.

Reference / interaction	Response	Gate	Stop	Target
AVP, first	Flow-flip full reentry	180	10	50
AVP, first	Pullback reassertion	600	10	30
AVP, second	Accepted break	180	10	30
TPO single print, second	Accepted break	180	10	30
Prior RTH value, first	Direct acceptance	180	10	30
TPO value, first	Flow-flip full reentry	600	10	50
TPO single print, second	One-second flow flip	30	15	45
TPO extreme, second	Accepted break	60	10	50
Developing value, first return	Migration flow flip	60	15	45
Completed-21 outer edge, first	Volume-window reversal	90	15	60

3 Distinct clocks, distinct hypotheses

Migration and long-memory profile rejection. Developing-value migration compares consecutive completed 30-minute snapshots of exact-contract VWAP, VPOC, VAL, and VAH. If all four rise, the prior VAH becomes a frozen support reference and only the first return from above is eligible. If all four fall, the prior VAL becomes resistance and only the first return from below is eligible. Mixed or unchanged directions, a first snapshot, and a contract mismatch do not qualify. A post-return flow flip satisfying Equation (2) must occur within 60 seconds. The hypothesis is that unanimous value relocation changes the interpretation of the old boundary.

The Completed-21 cell instead uses a completed 21-session profile's physical outer edge. At its first exact event-level touch, an outward 500-contract window followed by an inward window within 90 seconds triggers in the owner-edge direction: long at the lower edge, short at the upper edge. The intended rule uses an already available structure. A constructed source-input counterexample, however, shows that later receive timing can change earlier historical touch admission. This is an availability defect in the inspected pipeline, not a demonstrated count of affected historical trades.

Quote-anchor absorption on a volume clock. Displayed quote-regime anchors provide interaction levels for the V2000 absorption cell. After an eligible touch, the engine scans sequential completed 500-contract windows for aggressive flow pointing outward but no more than four ticks of outward open-to-extreme progress. A strictly later physical row must then move at least four ticks inward from the window's outward extreme. Heavy aggression accompanied by little price progress is an absorption *proxy*; trade-sampled quotes cannot identify hidden replenishment or prove a particular liquidity provider's behavior.

The earliest qualifying signal is detected before imposing the budget. Index physical events by the frozen lexicographic order (receive timestamp, source-file order, source-row number). Let j_0 be the touch row, j_* the signal row, and $\mathcal{V}$ the eligible nonduplicate rows retained before any integrity or identity termination. Then

$$B(j_*) = \sum_{\substack{j_0 < j \le j_* \\ j \in \mathcal{V}}} v_j \le 2{,}000. \tag{3}$$

The touch row is excluded; later physical rows with the same receive timestamp are included. The physical signal trade is counted in full, irrespective of window splitting. Exceeding the budget rejects the signal without deferred replacement. There is no wall-clock eligibility gate: 2,000 contracts may arrive quickly or slowly, and ordinary silence is not itself stale data. Stop/target is 12/20 points.

Impulse reversal and overnight outward continuation. The short-only impulse cell begins with a 500-contract footprint having at least 16 ticks of range, at least 12 ticks of displacement, and displacement-to-range efficiency of at least 0.70. At the second interaction, a flow-flip score of 25 must occur within 180 seconds. The historical implementation replays the full impulse population before retaining shorts for master ownership; it is not equivalent merely to disabling all long-side upstream state. Its stop/target is 15/45 points.

Overnight outward D2 uses the completed overnight VAL, VPOC, and VAH. It keeps the first four-tick break and a two-tick recapture within a 100-contract whole-row failure budget; there is no rearm. Despite the recapture, it trades in the *outward continuation* direction, with an 80-point stop and a 20-point target. This is a deliberately asymmetric payoff: many small wins can coexist with large individual losses. Its historical unresolved-position close marking requires separate qualification (Section 4.2).

Rolling-range failed-aggression reclaim. Rolling-30 uses the prior 30 completed one-minute RTH bars' high or low, not a reference recomputed with the signal bar; price differences below are NQ index points. For a lower-boundary long, let L_{30} be the prior low and O, H, L, C the signal bar. The defining geometry includes

$$4 \le L_{30} - L < 16, \qquad 1 \le C - L_{30} < 4, \qquad \frac{\min(O, C) - L}{H - L} \ge 0.50. \tag{4}$$

Outward signed-volume fraction must be at least 0.08. The upper-boundary short mirrors this geometry. The first signal per side uses 20/60-point exits. Excursion, reclaim, wick, and flow jointly describe failed aggression, not simply a close back inside a range.

4 VWAP release and execution economics

4.1 A frozen wall, not a moving target

For supplied completed bars $i < t$, define HLC3 price $h_i = (H_i + L_i + C_i)/3$ and exact deduplicated volume q_i. The research bridge computes

$$\mu_t = \frac{\sum_{i<t} q_i h_i}{\sum_{i<t} q_i}, \quad \sigma_t^2 = \frac{\sum_{i<t} q_i h_i^2}{\sum_{i<t} q_i} - \mu_t^2, \quad U_t = \mu_t + 2\sigma_t, \quad L_t = \mu_t - 2\sigma_t. \tag{5}$$

It clips numerical negative variance to zero and requires positive finite σ_t and at least 30 prior supplied bars. This is an HLC3 volume-weighted bar statistic, not transaction-price VWAP recomputed from every individual print. In the inspected bridge, eligible overnight rows can enter the prior-bar history; it is not necessarily an RTH-reset envelope.

A long release bar opens on or below U_t, closes at least $0.10\sigma_t$ above it, has an outward body, closes in the upper 30% of its range, and has aligned signed-volume fraction at least 0.05:

$$O_t \le U_t, \quad C_t \ge U_t + 0.10\sigma_t, \quad C_t > O_t, \quad \frac{C_t - L_t^{\text{bar}}}{H_t^{\text{bar}} - L_t^{\text{bar}}} \ge 0.70, \quad \frac{\Delta_t}{q_t} \ge 0.05. \tag{6}$$

Here superscript "bar" distinguishes a bar low/high from an envelope wall. The next supplied bar must be wholly above the frozen U_t, close no lower than the release close, and retain aligned flow of at least 0.05. The short condition mirrors these inequalities at the lower wall. Both bar opens must fall in [08:30,15:00) CT; exact one-minute adjacency is not enforced by this bridge. Confirmation availability supplies the signal time, and only the first successful episode per side is retained. Stop/target is 20/30 points.

Freezing the wall is economically important. Recomputing it after release would let the tested boundary move toward the breakout. Requiring an entire subsequent bar outside the old wall is a stronger persistence condition than asking only for another outside close.

4.2 Price, delay, and adverse execution

The shared historical convention uses one NQ, 3 milliseconds total decision/submission latency, and one adverse entry tick at an executable bid/ask observation. With side $s_i \in \{-1, +1\}$, fill e_i, and exit x_i, the basic one-contract accounting is

$$\pi_i = 20 s_i (x_i - e_i) - 2. \tag{7}$$

Targets are exact limits; a breached stop uses the executable quote plus one adverse tick. The entry event cannot also exit the trade, and stop takes priority in a same-event collision. Bid/ask execution

already matters independently of the extra tick. Consequently, a stop distance is not a guaranteed maximum dollar loss.

These shared constants do not imply identical lifecycles. For unresolved overnight D2 positions, the historical close mark retrospectively chooses the last quote through 15:00 CT and records that quote's earlier receive time. Only at cutoff is it known to be last; this is not an executable future liquidation, and the recorded timestamp can release capacity prematurely. The inspected VWAP scalar execution view lacks the event clock required for provider-age checks. It searches after signal plus 3 milliseconds without an explicit before-close entry bound and checks brackets before the close boundary, using quotes through 15:05. A quote at or after 15:00 may therefore bracket-resolve before censoring. We retain, disclose, and do not repair these historical behaviors in the reported PnL.

5 Ownership and measurement

5.1 Why attribution is not a portfolio-removal experiment

Let candidate i have signal time τ_i, recorded entry e_i^{time}, terminal time u_i, and component priority p_i. The historical ordering is lexicographic in signal time, entry time, priority, then candidate, signal, and trade identities. Priority groups are profile/flow, quote-anchor, an empty event-time reserve, short impulse, overnight outward, Rolling-30, and VWAP. The reserve is not a sixteenth signal.

For recorded busy-through time b, admission requires $\tau_i > b$; equality is blocked and rejected candidates are never deferred. An accepted trade updates b to its recorded terminal time. Denoting accepted ownership by A_i, the included daily PnL and sleeve attribution are

$$P_d = \sum_{i \in d} A_i \pi_i, \qquad C_{k,d} = \sum_{i \in d} A_i \mathbf{1}\{k(i) = k\} \pi_i, \qquad P_d = \sum_k C_{k,d}. \tag{8}$$

These are accounting identities for resolved, known included days. The marginal value of removing sleeve k is instead

$$M_k = P(\mathcal{S}) - P(\mathcal{S} \setminus \{k\}), \qquad M_k \neq \sum_d C_{k,d} \text{ in general.} \tag{9}$$

Fresh arbitration is needed: another sleeve may replace a removed owner and change later capacity intervals. Local simulated ownership can also suppress later source signals even when a candidate loses global ownership. The historical candidate population is filtered to filled/terminal outcomes, and entry time is not known at signal time. Absence of PnL ranking in the final arbiter is therefore not a proof of online equivalence.

5.2 Sample and statistical conventions

The explicit third-party `CME_Equity` calendar supplies 279 admitted sessions; official exchange confirmation is unavailable. Excluded dates are not invented as zero or unknown days. All inspected chronology is treated as development evidence. The exhaustive research trial inventory is unknown, so old partition labels do not create untouched holdouts or multiplicity-adjusted inference.

Within the included master scope, all 279 days are known. One unresolved Rolling-30 event on November 28, 2025 lies outside the arbitrated ledger. It is not inserted, valued at zero, or relabeled as an overlap rejection. "Complete" here cannot establish complete standalone opportunity history. Unknown parent days, if present, would remain null in every contribution and would break the

drawdown path. Empty sleeve observations justify a contribution zero only on a known-complete parent day.

Table 3: Included master economics, August 7, 2025–September 11, 2026.

Measure	Selected master
Admitted sessions (known / unknown)	279 / 0
Source / accepted / overlap-rejected	1,264 / 1,098 / 166
Resolved / censored trades	1,098 / 0
Resolved net / complete-known net (USD)	115,849 / 115,849
EV per accepted / resolved trade (USD)	105.51 / 105.51
Positive / negative net-trade sums (USD)	331,949 / 216,100
Profit factor / resolved-trade win rate	1.54 / 51.7%
Mean winning / mean losing trade magnitude (USD)	584.42 / 407.74
Green / red / zero / unknown sessions	172 / 96 / 11 / 0
Maximum daily-close drawdown (USD)	5,658
Longest observed underwater duration (closes)	22

Positive and negative trade sums use *net* trade outcomes, not pre-commission gross PnL. Drawdown is measured at admitted daily closes, not intraday. No return-bearing capital is specified; percentage returns, CAGR, and capital-based performance ratios are not reported.

6 The historical path: positive total, uneven experience

The master earns USD105.51 per accepted resolved trade, but the win rate is only 51.7%. Its mean winning trade is USD584.42 and mean loss magnitude is USD407.74. The selected aggregate therefore derives its positive result from both hit rate and payoff, not from near-universal wins. Across admitted days, mean PnL is USD415.23, median PnL USD398, and sample standard deviation USD1,201.95.

There are 172 green, 96 red, and 11 zero days: 61.6%, 34.4%, and 3.9% of all admitted sessions. The best known day, October 31, 2025, contributes USD4,088, or 3.5% of complete-known net. That relatively small single-day share does not establish broad opportunity independence: repeated flow primitives and shared NQ exposure connect the cells.

The empirical fifth percentile of daily PnL is $-\$1,768.10$; mean loss magnitude among the 14 days at or below that threshold is USD2,248.86. The longest observed underwater run is 22 closes. Drawdown depends on ordering and recovery, whereas the tail statistic summarizes individual poor days; neither establishes intraday account-risk compliance.

Longs account for 460 accepted trades and USD38,040; shorts for 638 and USD77,809. This is an exposure decomposition of accepted owners, not a long-versus-short randomized comparison. In particular, the short-only impulse rule makes directional participation structurally asymmetric.

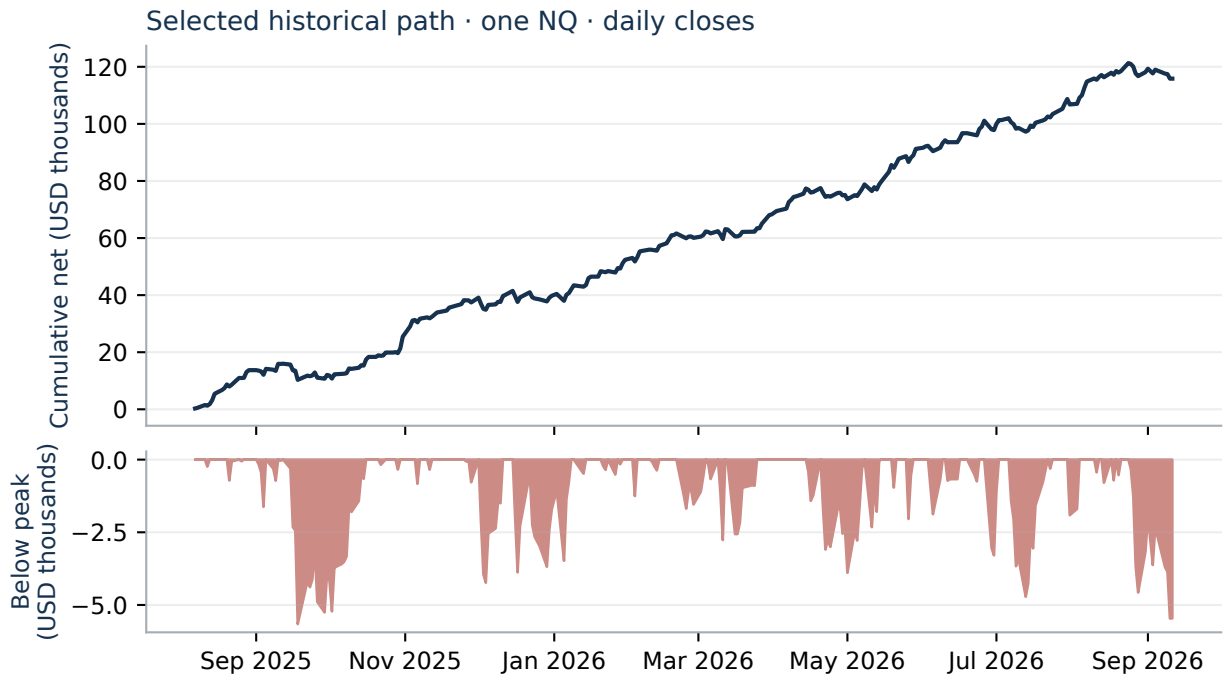


Figure 1: Cumulative net PnL and daily-close drawdown of the included ownership path. The vertical scale is dollars, not a return on an assumed account. The maximum observed daily-close drawdown is USD5,658; intraday adverse excursions are not identified by this figure.

7 Where the contribution comes from

Table 4: Accepted contribution by frozen signal cell. Src.: included source candidates; Own.: accepted trades; EV: USD per resolved trade. All fifteen cells remain shown, without selecting a favorable subset.

Signal cell	Src.	Own.	Net USD	EV	PF	Win
AVP first flow-flip reentry	15	13	3,424	263.4	3.19	38.5%
AVP pullback reassertion	15	13	2,039	156.8	2.32	46.2%
AVP second acceptance	30	27	1,621	60.0	1.43	33.3%
TPO single-print acceptance	34	22	1,746	79.4	1.57	36.4%
Prior-value direct acceptance	84	71	1,823	25.7	1.17	29.6%
TPO value reentry	98	77	6,771	87.9	1.56	24.7%
TPO single-print flow flip	65	57	4,741	83.2	1.39	33.3%
TPO extreme acceptance	13	6	2,373	395.5	4.82	50.0%
Developing-value migration	39	39	10,637	272.7	2.72	48.7%
Completed-21 outer reversal	36	28	4,874	174.1	1.82	32.1%
Quote-anchor absorption	117	111	3,508	31.6	1.24	45.0%
Short-only impulse flip	194	158	12,334	78.1	1.37	32.3%
Overnight outward D2	356	318	27,464	86.4	1.35	84.3%
Rolling-30 reclaim	91	87	17,241	198.2	1.85	42.5%
VWAP wall release	77	71	15,253	214.8	2.38	62.0%

Overnight outward D2 is the largest single-cell net contributor at USD27,464, approximately 23.7% of total net, across 318 accepted trades. Rolling-30 contributes USD17,241 and VWAP wall release USD15,253. Together, these three cells account for approximately 51.8% of net, despite markedly different triggers and payoff shapes. The ten profile/flow cells jointly contribute USD40,049; the

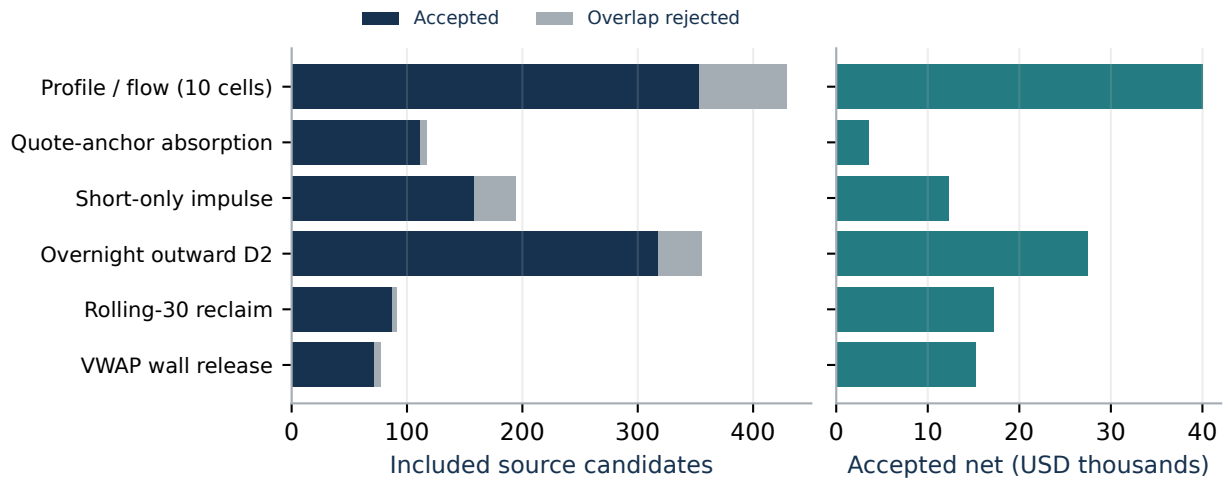


Figure 2: Capacity use and accepted PnL by source group, in policy order. Counts on the left include recorded overlap rejections. Net contributions on the right partition the master; they are not standalone returns or incremental portfolio benefits.

aggregation is useful, but it should not be mistaken for ten independent sources of edge.

Sparse cells illustrate why ranking on profit factor is inappropriate. TPO extreme acceptance has a 4.82 profit factor, but only six accepted trades. The developing-value migration cell contributes USD10,637 from 39 trades, whereas the quote-anchor cell contributes USD3,508 from 111. Those contrasts describe the inspected sample, not a reliable ranking of prospective performance. Every attribution in Table 4 is conditional on the existing roster and recorded capacity decisions.

8 Payoff geometry, standalone evidence, and time variation

8.1 Equal contract count is not equal risk

Ignoring fees, slippage, gaps, and non-bracket exits, a binary stop/target payoff with stop S and target T breaks even at

$$p^* = \frac{S}{S+T}, \qquad \mathbb{E}[\pi] \approx 20\{pT - (1-p)S\} - c. \tag{10}$$

This is a geometry illustration, not a replacement for recorded fills. Overnight D2's 80/20 geometry implies an 80% costless break-even hit rate. Its observed 84.3% win rate is thus not comparable at face value with the 32.3% short-impulse hit rate under 15/45 geometry, whose corresponding threshold is 25%. Actual net expectancy depends on slippage, gaps, commissions, and close handling.

The D2 accepted-contribution path has an USD8,483 maximum daily-close drawdown, exceeding the master's USD5,658. This is a difference between two observed accounting paths, not proof of a diversification benefit against an independently executed D2 benchmark. Similarly, one contract with an 80-point stop has USD1,600 nominal stop-distance exposure before execution costs, versus USD200 for a 10-point stop. Counting contracts alone obscures an eightfold difference in nominal geometry.

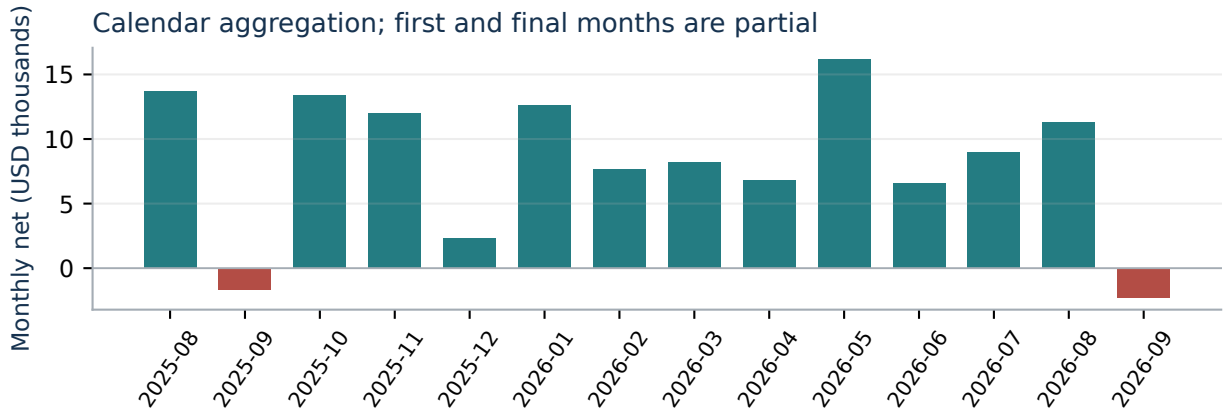


Figure 3: Monthly net PnL of the master. August 2025 begins on the 7th; September 2026 ends on the 11th. Twelve of fourteen represented months are positive, but unequal and partial month lengths preclude interpreting the bars as equal-exposure trials.

8.2 A genuine standalone coordinate answers a different question

Table 5: VWAP wall release under two supplied ownership scopes. The separate standalone path is not added to the master.

VWAP wall-release measure	Master contribution	Standalone
Accepted trades	71	77
Resolved net (USD)	15,253	15,826
EV / resolved trade (USD)	214.83	205.53
Profit factor	2.38	2.29
Daily-close drawdown (USD)	1,050	1,050

The standalone VWAP coordinate owns 77 trades; the master owns 71 of its source candidates. Standalone net exceeds accepted contribution by USD573, while its per-trade expectancy is lower. This shows why "standalone minus contribution" and "value added to the portfolio" must not be conflated: the latter requires replaying replacement owners and all resulting capacity intervals. Both supplied coordinates inherit the historical implementation qualifications; "standalone" describes ownership scope, not stronger causal validation.

9 Fixed recency economics, without window selection

All three predeclared trailing windows end at September 11, 2026. They are the last 5, 10, and 30 *admitted* sessions, not calendar days or the last trades. No window is selected for favorable performance. Exact date lists and retained daily states are supplied in the aggregate companion described in Appendix A, for all fifteen contribution scopes, the master, and standalone VWAP.

Table 6: Required trailing-session diagnostics for the master, ending September 11, 2026. Monetary quantities are USD. Shares use the full fixed-window session count.

Measure	L5	L10	L30
Start (2026)	09-04	08-28	07-31
Source / accepted / resolved / censored	12 / 11 / 11 / 0	32 / 31 / 31 / 0	136 / 125 / 125 / 0
Resolved net (USD)	-1,837	-1,777	7,130
Complete-known net (USD)	-1,837	-1,777	7,130
EV / accepted event (USD)	-167.00	-57.32	57.04
EV / resolved trade (USD)	-167.00	-57.32	57.04
Positive net-trade sum (USD)	2,692	7,578	35,490
Negative net-trade magnitude (USD)	4,529	9,355	28,360
Profit factor	0.59	0.81	1.25
Known-segment drawdown (USD)	3,133	3,488	5,463
Green / red / zero / unknown days	1 / 3 / 1 / 0	3 / 6 / 1 / 0	15 / 14 / 1 / 0
Green share / red share	20.0% / 60.0%	30.0% / 60.0%	50.0% / 46.7%
Zero share / unknown share	20.0% / 0.0%	10.0% / 0.0%	3.3% / 0.0%
Wins / losses / zero-PnL trades	4 / 7 / 0	11 / 20 / 0	55 / 70 / 0
Win-rate support / win rate	11 / 36.4%	31 / 35.5%	125 / 44.0%
Best known day (2026)	09-04	08-31	08-24
Best known day (USD)	1,296	1,400	2,794
Best day / complete-known net	-70.5%	-78.8%	39.2%

The last five sessions lose USD1,837 and the last ten lose USD1,777. Their profit factors, 0.59 and 0.81, contrast with the full-history 1.54. The thirty-session window remains positive at USD7,130, but its USD5,463 daily-close drawdown approaches the full-history maximum. A positive longer-window total therefore does not erase recent weakness or the experience of traversing that path.

The windows overlap and are not independent tests. L5's one green day contributes USD1,296, even though total net is negative; its best-day share is consequently negative. L10 has the same interpretive issue. A negative or above-one share is valid arithmetic, not a reason to suppress the row or rename it a stability measure.

Resolved subtotal and complete-known-day PnL coincide here but remain separate fields. An unresolved accepted trade can leave the day null even when other trades resolve. Unknown days retain their denominator weight, cannot become green/red/zero days, and interrupt drawdown paths.

10 Cost sensitivity and what the statistics cannot establish

10.1 A transparent incremental-cost calculation

Let c be an additional dollar charge per resolved trade, on top of the historical net ledger. Holding outcomes and ownership fixed gives

$$\Pi(c) = \Pi(0) - Nc, \qquad \overline{\pi}(c) = \overline{\pi}(0) - c, \qquad N = 1{,}098. \tag{11}$$

This is an accounting stress, not a latency or market-impact replay. The fixed charges are USD0, 5, 10, and 20; no favorable cost case is selected. An additional USD20 per resolved trade reduces total net to USD93,889 and expectancy to USD85.51. The full-history arithmetic break-even extra charge is USD105.51 per trade. That number is not an executable slippage allowance: worsening fills can alter stops, opportunities, ownership, and censoring nonlinearly.

Table 7: Per-resolved-trade additional charges; all PnLs remain after the original modeled costs.

Extra USD / trade	Net USD	EV USD	PF
0	115,849	105.51	1.54
5	110,359	100.51	1.50
10	104,869	95.51	1.47
20	93,889	85.51	1.41

10.2 Dependence and historical selection

Order-flow persistence and heterogeneous responses make independent-trade inference implausible without additional evidence. Kyle [1] and Hasbrouck [2] motivate the connection between trades, information, and price adjustment; Cont, Kukanov, and Stoikov [3] emphasize order-book events and price impact. These studies do not validate the thresholds here. In particular, signed traded volume sampled with BBO is not the same measure as full order-book imbalance.

As a descriptive sampling diagnostic, a five-session noncircular moving-block bootstrap (1,000 resamples, fixed seed zero) gives a 95% interval of USD298.45–555.85 for mean daily PnL. A Bartlett-kernel Newey–West calculation with four lags gives USD274.32–556.14. Both are conditional on the supplied selected path and depend on stability and dependence assumptions. They do not adjust for an unknown number of tried strategies, source defects, or choice of portfolio. Following White [4], a favorable selected mean cannot be treated as an unadjusted confirmatory discovery. Lo [5] and Newey and West [6] likewise caution against dependence-insensitive interpretation of performance statistics.

11 Conclusion and the next discriminating tests

The selected portfolio combines acceptance, rejection, absorption, failed excursions, and VWAP-envelope persistence. Volume windows, one-second flow, elapsed-time gates, and completed bars are distinct information clocks, not interchangeable indicator settings.

The recorded one-position path earns USD115,849 with a 1.54 profit factor, but L5 and L10 are negative and some high cell-level ratios have very small support. This is a selected ownership path, not fifteen independent edges. Conditional contributions reconcile; marginal benefits remain unmeasured.

Three tests remain decisive: quantify the actual incidence and economic effects of the documented timing defects; replay sleeve removals with fresh arbitration and local states; and compare the frozen roster with simpler mechanism and exposure benchmarks on unseen chronology, retaining the trial inventory and synchronized execution stresses. A favorable cumulative curve replaces none of these tests.

This paper is historical research, not a recommendation to trade.

References

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A Exact recency membership and aggregate companion

The companion `aggregate_economics.json` retains exact ordered L5/L10/L30 dates, starts, ends, cutoff, and daily aggregate states for all seventeen scopes. It includes full-history, monthly, directional, contribution, standalone VWAP, and cost summaries, but no per-trade rows. L5 comprises September 4, 8, 9, 10, and 11, 2026; the longer exact lists follow the same supplied calendar, not inferred weekdays.

EV is null at zero support. Win rate excludes zero-PnL trades. Profit factor uses positive and negative *net* trade sums; gains without losses have positive-infinity status, while no nonzero outcomes are undefined. JSON encodes nonfinite values as null with explicit status. Unknown days reset cumulative PnL and its drawdown peak. Best-day share is null at zero complete-known net. The shared recency engine reconciles daily/event totals; this establishes output consistency, not source-to-signal causality.